Name: Neeraj Pandey

Roll No. 2401073

Class: T21

EXPERIMENT NO. 08

Aim: CI/CD Pipeline to host a website on AWS

Theory:

CI/CD (Continuous Integration and Continuous Deployment) is a software development practice that automates the process of integrating code changes, testing applications, and deploying them to a production environment. It helps developers deliver software faster, reduce human errors, and maintain application quality.

Continuous Integration (CI) is the process in which developers frequently push their code to a shared repository, such as GitHub. Whenever new code is pushed, an automated workflow is triggered to build the application and run tests to identify errors.

Continuous Deployment (CD) is the process of automatically deploying successfully tested code to a hosting environment. This ensures that the latest version of the website is available to users without requiring manual deployment for every change.

AWS Services Used

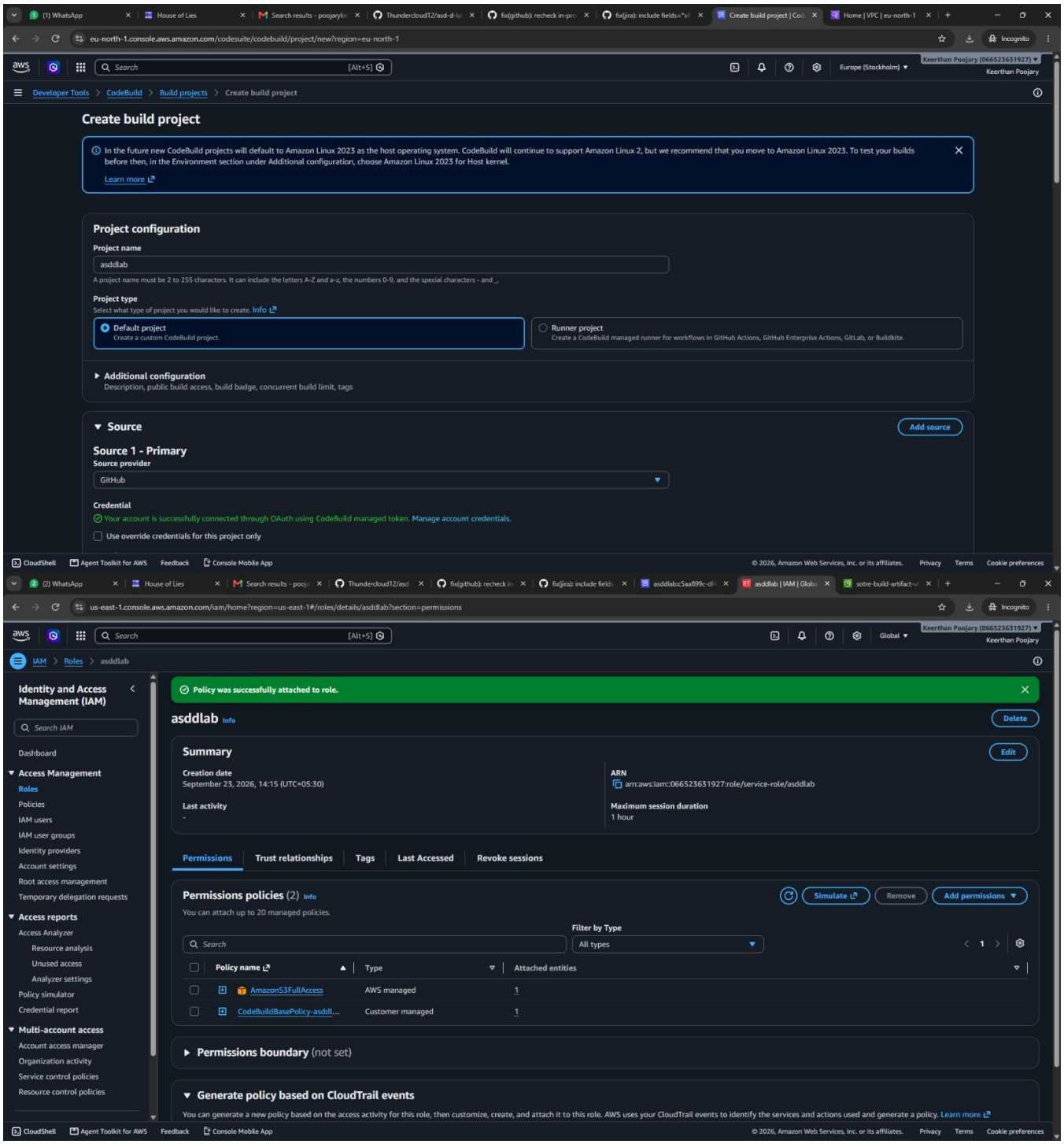
1. Amazon Web Services (AWS): A cloud computing platform that provides infrastructure and services for hosting websites and applications.
2. Amazon S3: A storage service that can host static websites containing HTML, CSS, JavaScript, and other static files.
3. GitHub: A version control platform used to store source code and track changes.

4. GitHub Actions: An automation tool used to create CI/CD workflows that build, test, and deploy the website.
5. AWS IAM: Identity and Access Management controls access to AWS resources and allows the deployment workflow to receive only the permissions it needs.

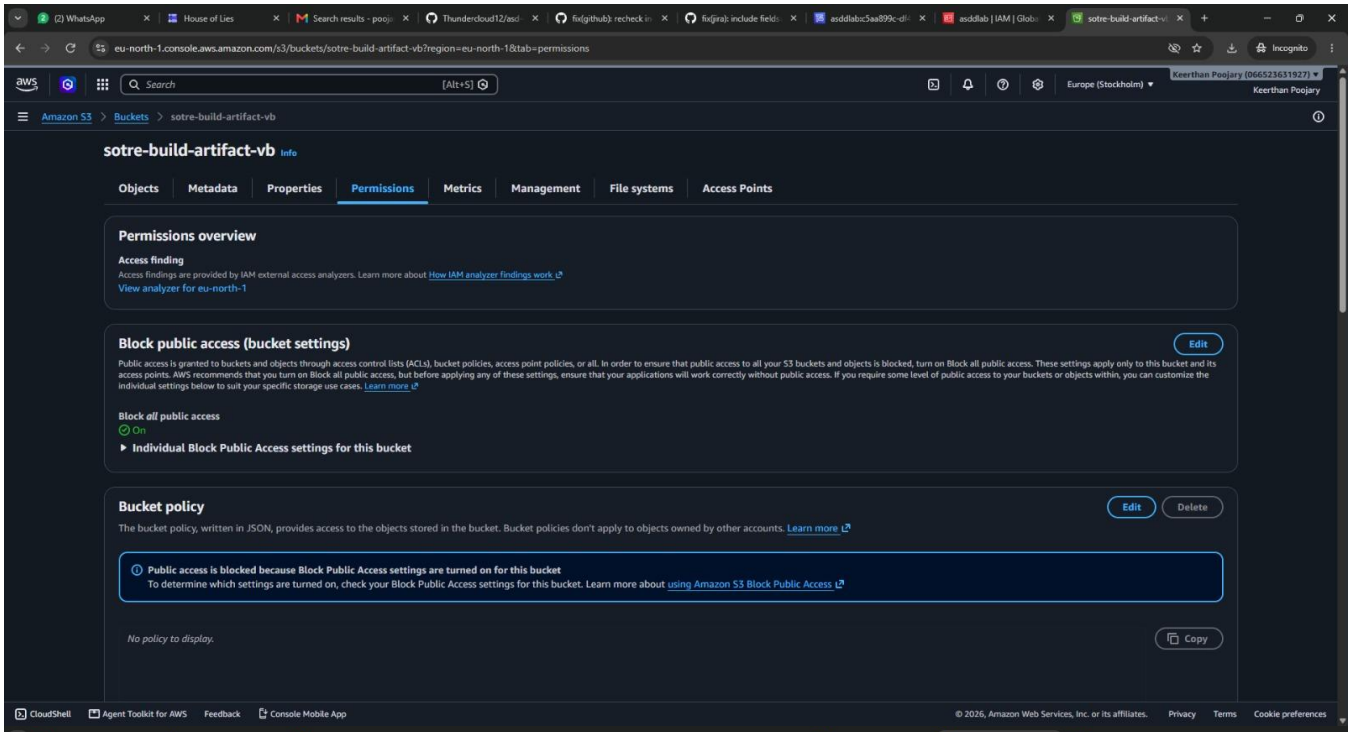
Working of the CI/CD Pipeline

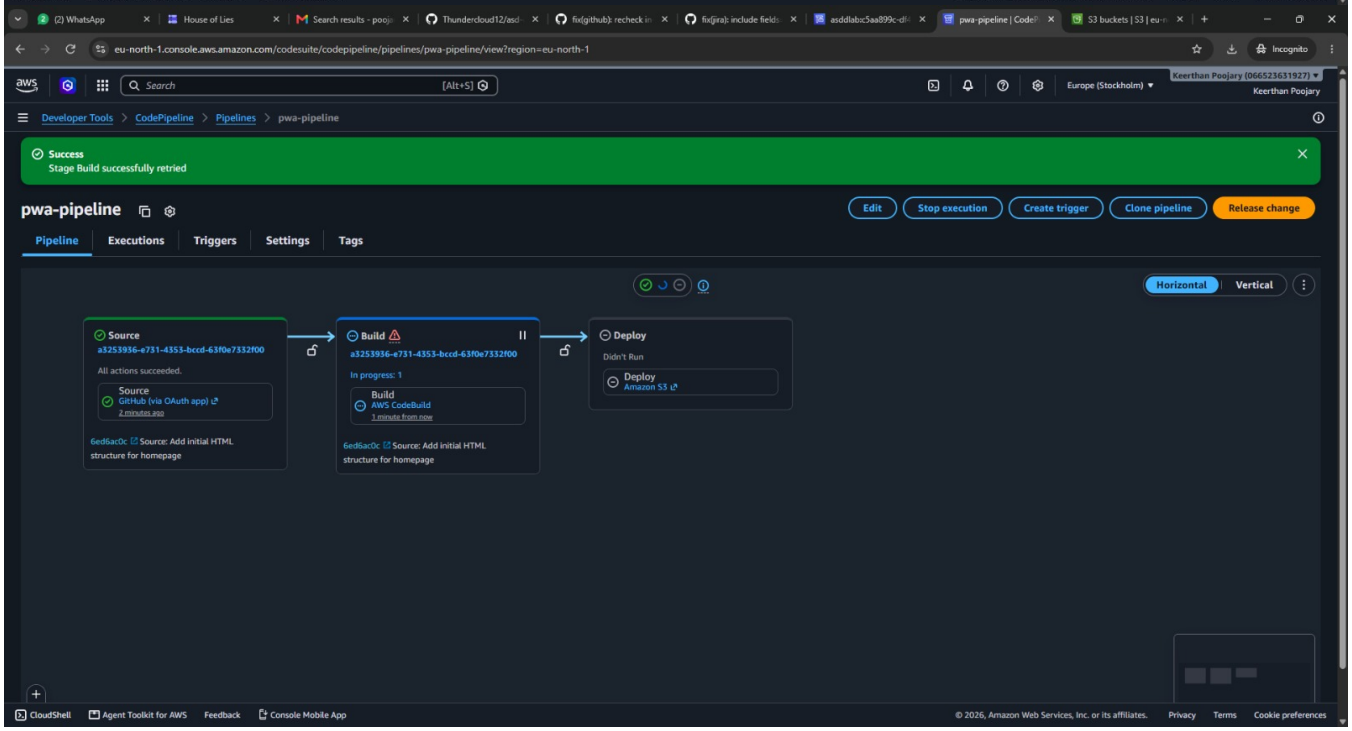
The CI/CD pipeline follows these steps:

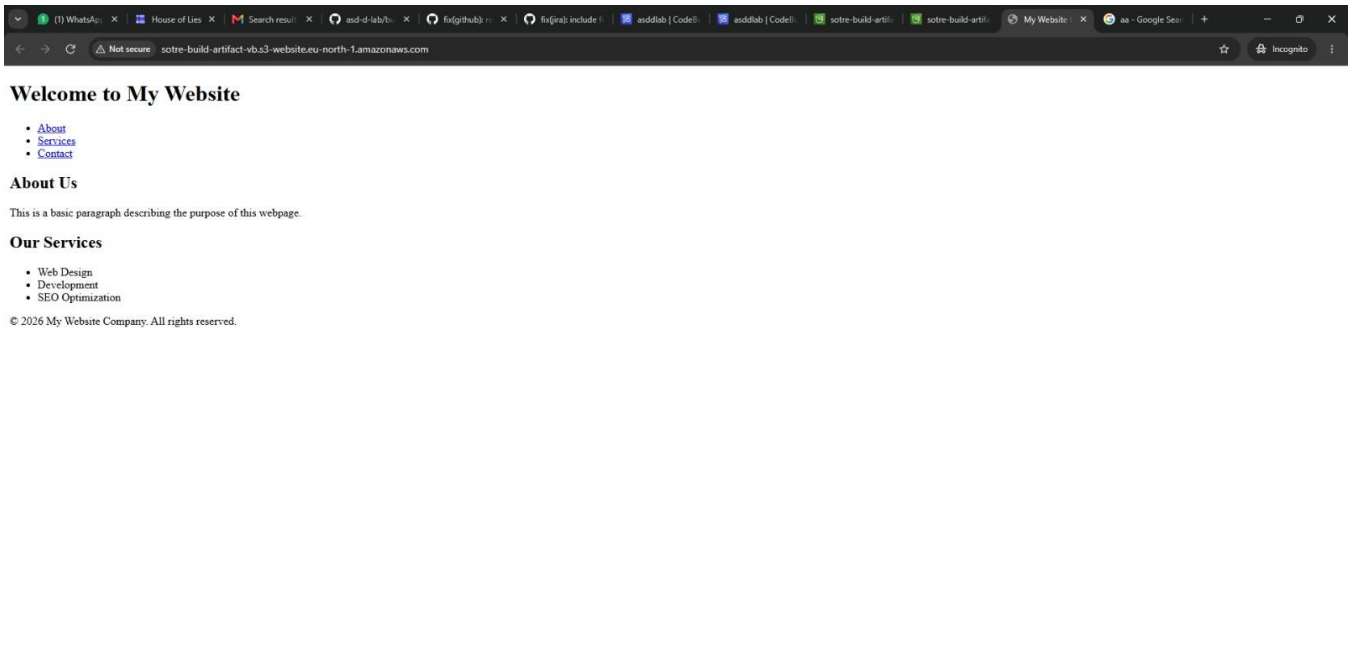
1. Code Development: The developer creates or modifies the website source code.
2. Code Repository: The source code is pushed to a GitHub repository.
3. Continuous Integration: GitHub Actions automatically triggers a workflow when new code is pushed.
4. Build and Testing: The workflow installs dependencies, builds the website if necessary, and runs automated tests.
5. Continuous Deployment: If the build and tests succeed, the workflow uploads the website files to an Amazon S3 bucket.
6. Website Hosting: Amazon S3 serves the deployed static website to users through its website endpoint. For production websites, Amazon CloudFront can provide HTTPS and content delivery.

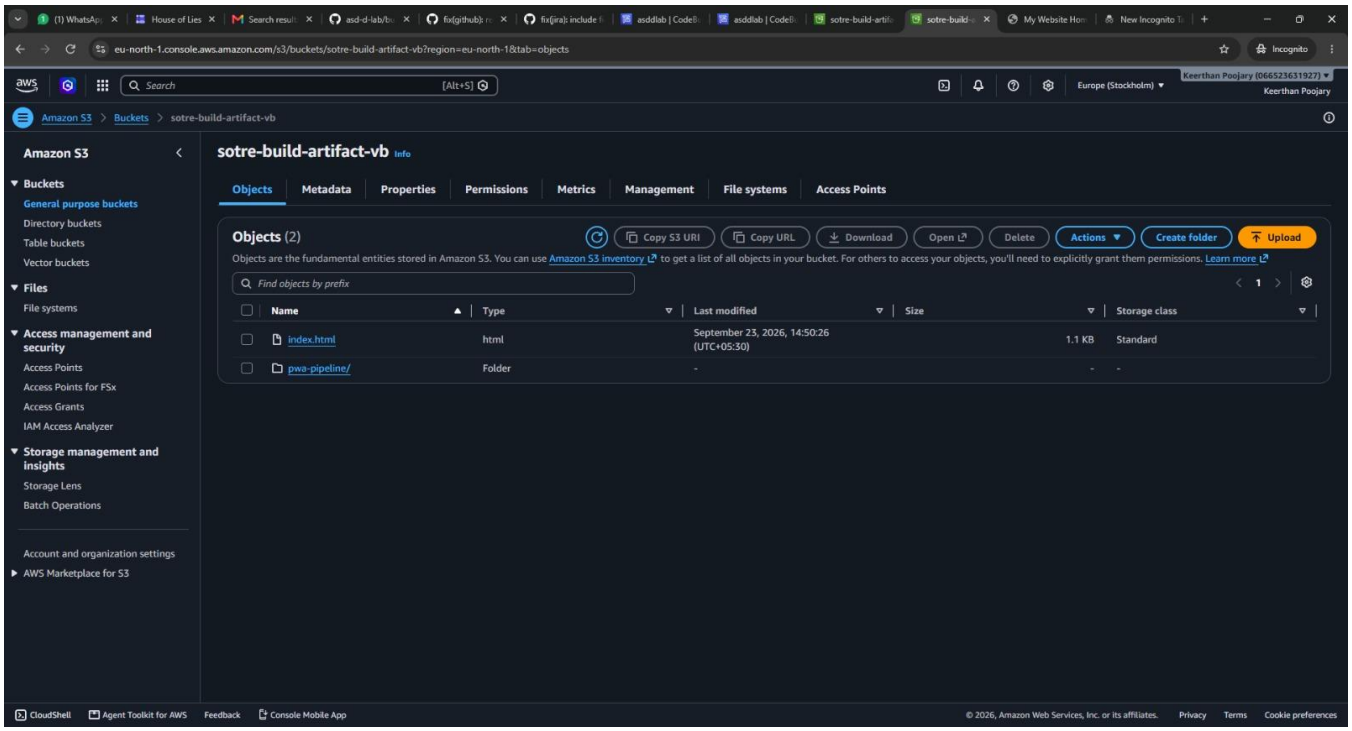

eu-north-1.console.aws.amazon.com/codesuite/codebuild/project/new?region=eu-north-1
Developer Tools > CodeBuild > Build projects > Create build project
Create build project
In the future new CodeBuild projects will default to Amazon Linux 2023 as the host operating system. CodeBuild will continue to support Amazon Linux 2, but we recommend that you move to Amazon Linux 2023. To test your builds before then, in the Environment section under Additional configuration, choose Amazon Linux 2023 for Host kernel.
Learn more
Project configuration
Project name
asddlab
A project name must be 2 to 255 characters. It can include the letters A-Z and a-z, the numbers 0-9, and the special characters - and _.
Project type
Select what type of project you would like to create. Info
Default project
Create a custom CodeBuild project.
Runner project
Create a CodeBuild managed runner for workflows in GitHub Actions, GitHub Enterprise Actions, GitLab, or Buildkite.
Additional configuration
Description, public build access, build badge, concurrent build limit, tags
Source
Add source
Source 1 - Primary
Source provider
GitHub
Credential
Your account is successfully connected through OAuth using CodeBuild managed token. Manage account credentials.
Use override credentials for this project only
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

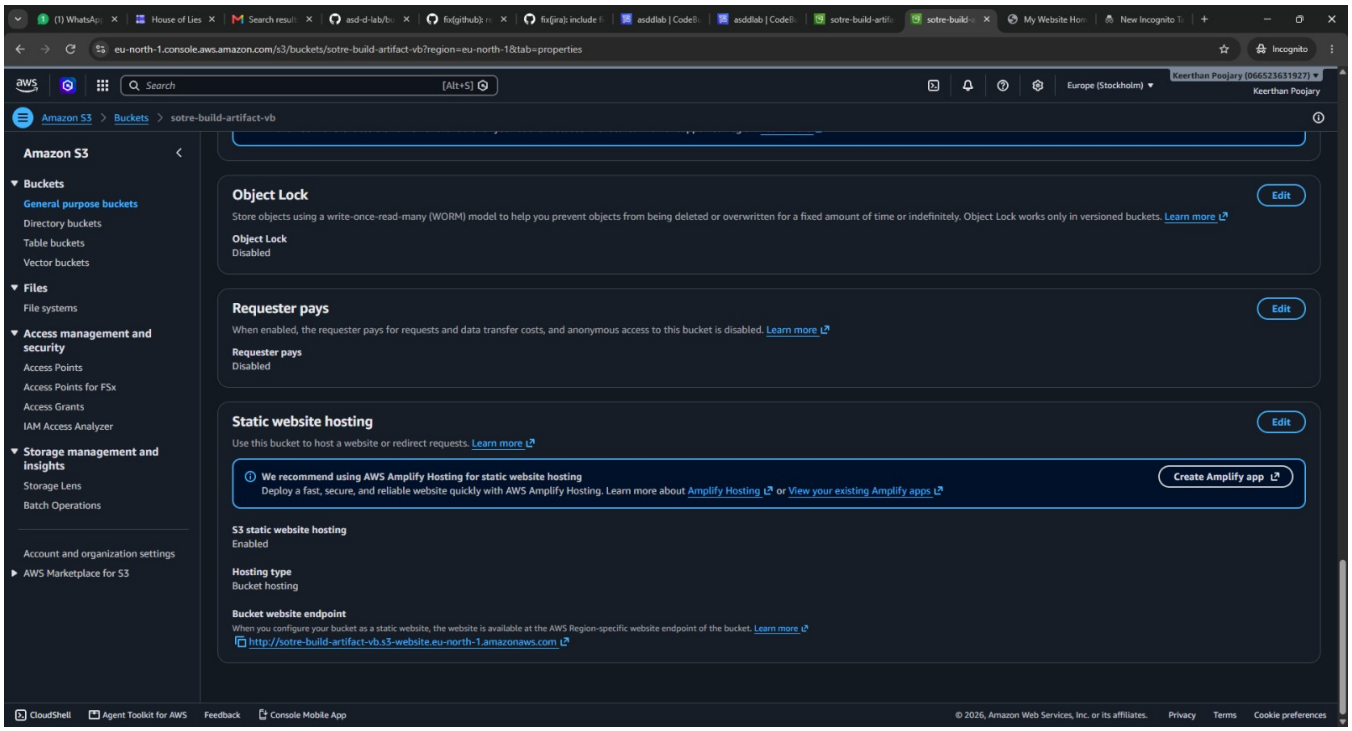
us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/roles/details/asddlab?section=permissions
IAM > Roles > asddlab
Identity and Access Management (IAM)
Search IAM
Dashboard
Access Management
Roles
Policies
IAM users
IAM user groups
Identity providers
Account settings
Root access management
Temporary delegation requests
Access reports
Access Analyzer
Resource analysis
Unused access
Analyzer settings
Policy simulator
Credential report
Multi-account access
Account access manager
Organization activity
Service control policies
Resource control policies
Policy was successfully attached to role.
asddlab Info
Delete
Summary
Edit
Creation date
September 23, 2026, 14:15 (UTC+05:30)
ARN
arn:aws:iam::066523631927:role/service-role/asddlab
Last activity
-
Maximum session duration
1 hour
Permissions
Trust relationships
Tags
Last Accessed
Revoke sessions
Permissions policies (2) Info
You can attach up to 20 managed policies.
Simulate
Remove
Add permissions
Filter by Type
Search
All types
Policy name
Type
Attached entities
AmazonS3FullAccess
AWS managed
1
CodeBuildBasePolicy-asddl...
Customer managed
1
Permissions boundary (not set)
Generate policy based on CloudTrail events
You can generate a new policy based on the access activity for this role, then customize, create, and attach it to this role. AWS uses your CloudTrail events to identify the services and actions used and generate a policy. Learn more
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


eu-north-1.console.aws.amazon.com/s3/buckets/sotre-build-artifact-vb?region=eu-north-1&tab=permissions
Amazon S3 > Buckets > sotre-build-artifact-vb
sotre-build-artifact-vb Info
Objects
Metadata
Properties
Permissions
Metrics
Management
File systems
Access Points
Permissions overview
Access finding
Access findings are provided by IAM external access analyzers. Learn more about How IAM analyzer findings work
View analyzer for eu-north-1
Block public access (bucket settings)
Edit
Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. Learn more
Block all public access
On
Individual Block Public Access settings for this bucket
Bucket policy
Edit
Delete
The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. Learn more
Public access is blocked because Block Public Access settings are turned on for this bucket
To determine which settings are turned on, check your Block Public Access settings for this bucket. Learn more about using Amazon S3 Block Public Access
No policy to display.
Copy
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


eu-north-1.console.aws.amazon.com/codesuite/codepipeline/pipelines/pwa-pipeline/view?region=eu-north-1
Developer Tools > CodePipeline > Pipelines > pwa-pipeline
Success
Stage Build successfully retried
pwa-pipeline
Edit
Stop execution
Create trigger
Clone pipeline
Release change
Pipeline
Executions
Triggers
Settings
Tags
Horizontal
Vertical
Source
a3253936-e731-4353-bccd-63f0e7332f00
All actions succeeded.
Source
GitHub (via OAuth app)
2 minutes ago
6ed6ac0c Source: Add initial HTML structure for homepage
Build
a3253936-e731-4353-bccd-63f0e7332f00
In progress: 1
Build
AWS CodeBuild
1 minute from now
6ed6ac0c Source: Add initial HTML structure for homepage
Deploy
Didn't Run
Deploy
Amazon S3
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Not secure
sotre-build-artifact-vb.s3-website.eu-north-1.amazonaws.com
Incognito
Welcome to My Website
About
Services
Contact
About Us
This is a basic paragraph describing the purpose of this webpage.
Our Services
Web Design
Development
SEO Optimization
© 2026 My Website Company. All rights reserved.


eu-north-1.console.aws.amazon.com/s3/buckets/sotre-build-artifact-vb?region=eu-north-1&tab=objects
Incognito
Search
[Alt+S]
Europe (Stockholm)
Keerthan Poojary (066523631927)
Keerthan Poojary
Amazon S3 > Buckets > sotre-build-artifact-vb
Amazon S3
Buckets
General purpose buckets
Directory buckets
Table buckets
Vector buckets
Files
File systems
Access management and security
Access Points
Access Points for FSx
Access Grants
IAM Access Analyzer
Storage management and insights
Storage Lens
Batch Operations
Account and organization settings
AWS Marketplace for S3
sotre-build-artifact-vb Info
Objects
Metadata
Properties
Permissions
Metrics
Management
File systems
Access Points
Objects (2)
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Find objects by prefix
Name
Type
Last modified
Size
Storage class
index.html
html
September 23, 2026, 14:50:26 (UTC+05:30)
1.1 KB
Standard
pwa-pipeline/
Folder
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences



Conclusion:

A CI/CD pipeline using GitHub Actions and AWS automates the process of building, testing, and deploying a website. By integrating version control with cloud hosting, developers can publish updates quickly and reliably. This approach improves development efficiency, reduces deployment errors, and demonstrates the practical use of cloud computing and DevOps principles